



CONSUMER SHOPPING BEHAVIOUR ANALYSIS

Stage 0 | Lawal Mayowa Bryant

INTRODUCTION

This project analyses consumer shopping behaviour to uncover trends in purchases, discounts, product categories, and customer preferences using Excel.

DATASET OVERVIEW

- Dataset Name: Consumer Shopping Behaviour

- Source: [Kaggle Link](https://www.kaggle.com/datasets/zeesolver/consumer-behavior-and-shopping-habits-dataset?select=shopping_behavior_updated.csv)

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- Rows / Columns: 3,901 rows × 20 columns

- Key Features: Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount, Discount Applied, Promo Code Used, Shipping Type, Subscription Status, etc.

RESEARCH QUESTIONS

1. How does the age group affect average purchase amount?
2. Do subscribed customers spend more than non-subscribers?
3. Which product category generates the highest total sales?
4. Do discounts or promo codes increase the purchase amount?

DATA CLEANING SUMMARY

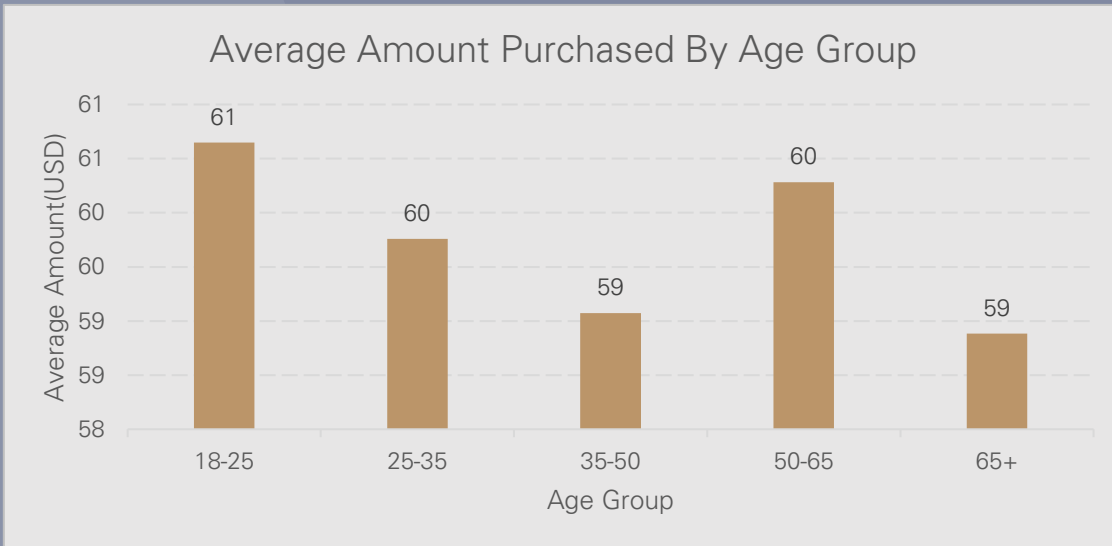
Steps Taken to Clean and Prepare the Data:

- Checked for missing values using Excel's COUNTBLANK formula and replaced blanks with appropriate values (e.g., "Unknown" or averages).
- Identified and corrected inconsistent entries (e.g., "Yes" vs "yes") using Find & Replace.
- Fixed spelling errors and standardized text formatting (capitalization, spacing).
- Handled outliers by reviewing extreme purchase values and ensuring they were valid data points.
- Added a new Age Group column to categorize customers for easier analysis.

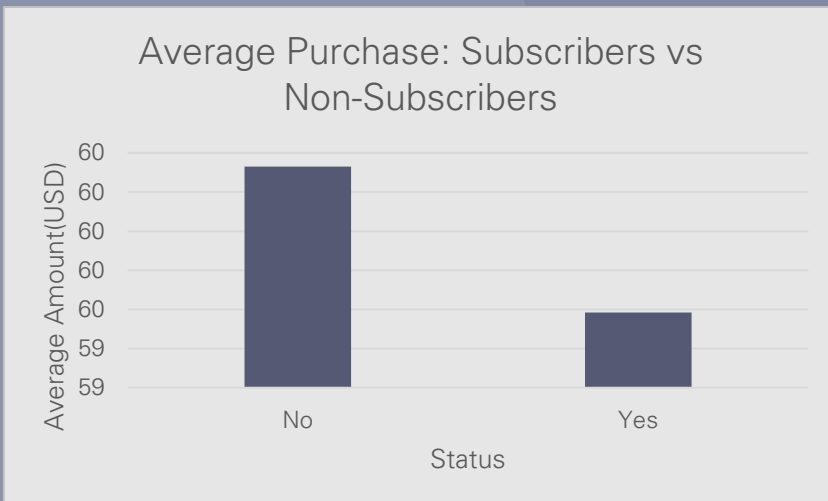
Tools and Techniques Used:

- Excel functions: COUNTBLANK, TRIM, PROPER, and IF
- Find & Replace for consistency
- PivotTables and Charts for analysis and visualization

ANALYSIS – PART 1



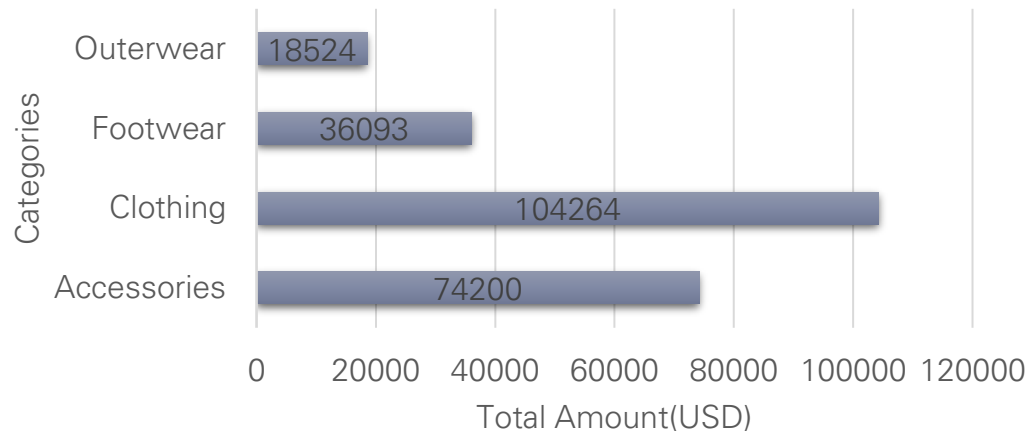
- Title:** Average Purchase by Age Group
- Insight:** Average purchase amount is similar across all age groups (~\$60), showing no single age group dominates spending.
- Conclusion:** Age group has minimal impact on average purchase; other factors may influence spending more.



- Title:** Average Purchase: Subscribers vs Non-Subscribers
- Insight:** Subscribers spend slightly less (\$59) than non-subscribers (\$60).
- Conclusion:** Subscription status does not significantly affect spending; marketing should consider other drivers.

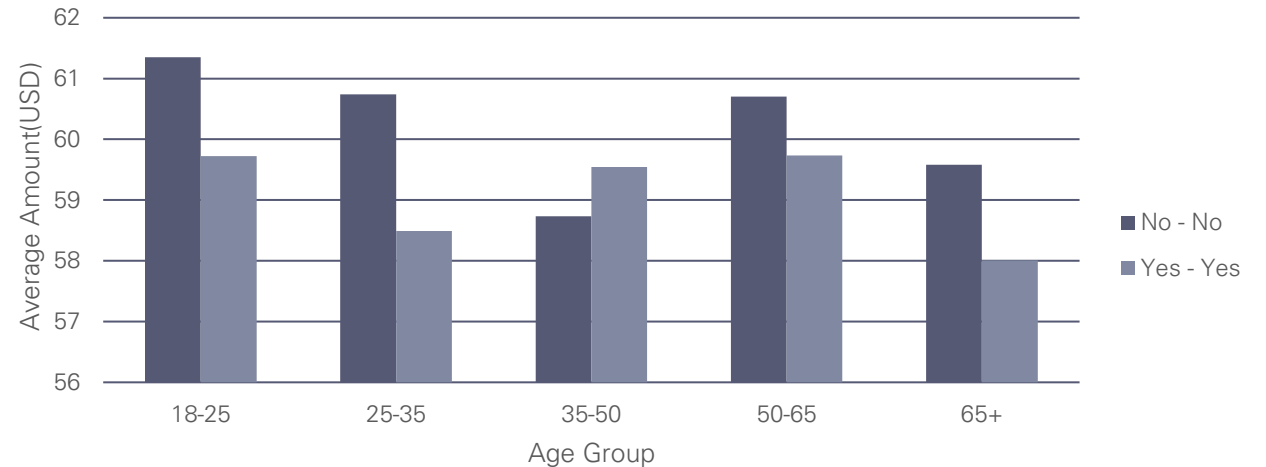
ANALYSIS – PART 2

Top-Selling Product Categories



- Title:** Total Sales by Product Category
- Insight:** Clothing generates the highest total sales (\$104,264), followed by Accessories. Footwear and Outerwear contribute much less.
- Conclusion:** Focus marketing and inventory efforts on Clothing and Accessories to maximize revenue.

Impact of Discounts & Promo Codes



- Title:** Impact of Discounts & Promo Codes
- Insight:** Customers with discounts or promo codes spend slightly more (\$59–\$61), indicating promotions encourage purchases.
- Conclusion:** Promotional strategies have a small but positive effect on spending; consider targeted discounts.

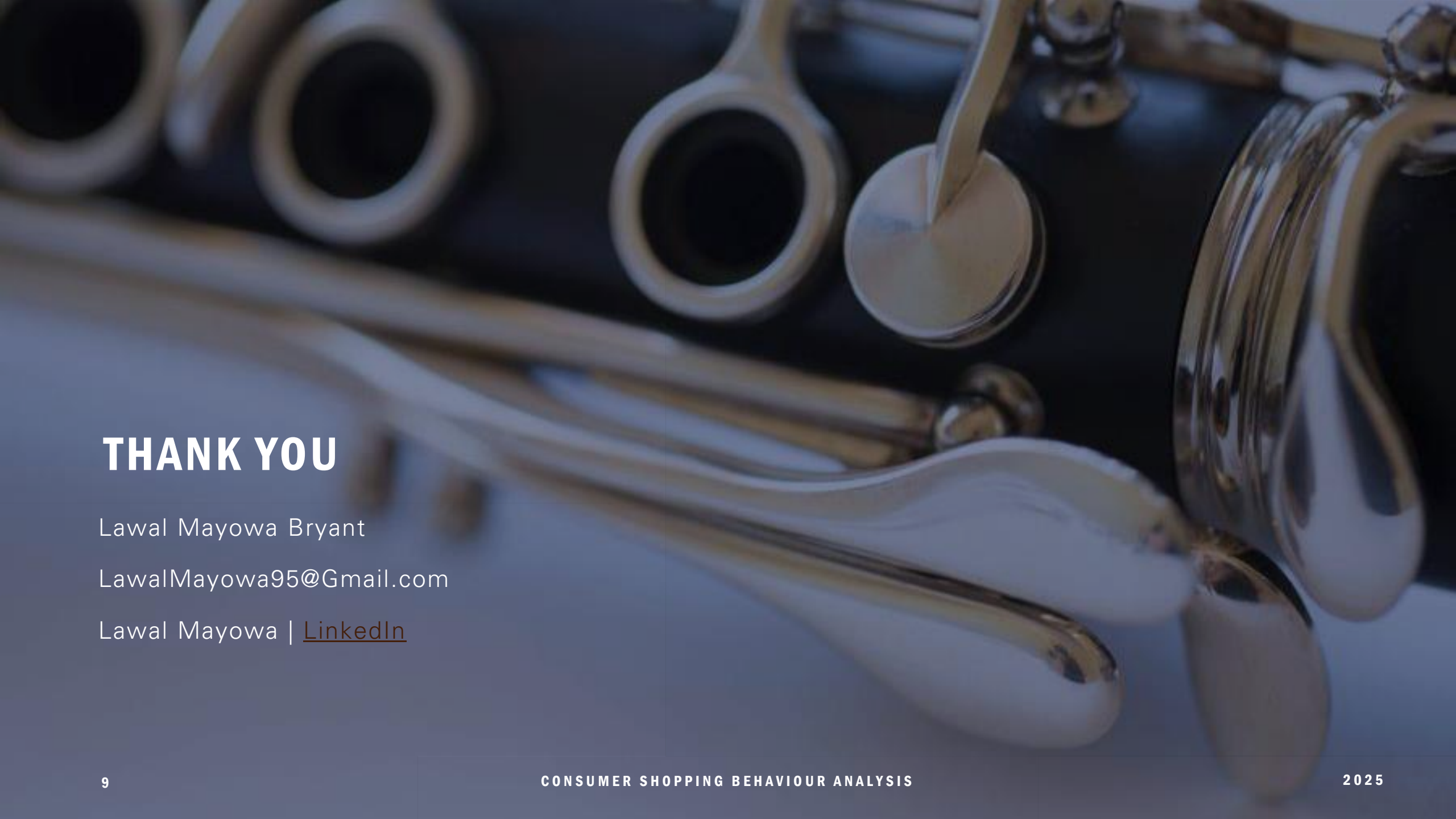
LIMITATIONS & FUTURE WORK

Limitations

- ❑ The dataset is **relatively small**, so the insights may not fully represent all shopping behaviours.
- ❑ The dataset **doesn't specify a timeline**, preventing analysis of trends or seasonal patterns.
- ❑ Some records are **incomplete or unevenly distributed**, possibly biasing results for certain age groups or locations.

Future Work

- ❑ **Collect more data** across multiple time periods to capture seasonal and yearly trends.
- ❑ **Integrate additional variables** such as income level or marketing spend to deepen analysis.
- ❑ **Use visualization tools** like Power BI or Tableau for more dynamic dashboards.
- ❑ Explore **predictive analytics** later to forecast customer behaviour or sales trends.



THANK YOU

Lawal Mayowa Bryant

LawalMayowa95@Gmail.com

Lawal Mayowa | [LinkedIn](#)